



United International University

Department of Computer Science and Engineering

SOC 2101: Society, Environment and Engineering Ethics

Final Exam. : Spring 2025

Total Marks: 25 Time: 1 hour 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the 5 questions. Numbers to the right of the questions denote their marks.

1. Mr. Z is a senior developer at a robotics company specializing in AI-powered drones initially developed for agricultural automation tasks such as crop monitoring and pesticide spraying. Recently, the company signed a lucrative contract with a defense agency to adapt the same drone software for military surveillance and tactical operations. Although the drones were never designed or tested for high-risk environments, the company leadership insisted on repurposing the software without any significant modifications. Mr. Z was aware that under certain conditions—such as fog, low lighting, or sudden movement—the drones could misclassify targets and that such errors could have life-threatening consequences in military contexts. When he raised these concerns with his superiors, he was dismissed and advised to focus on delivering the project quickly. Fearing job insecurity and lured by the promise of a performance bonus, Mr. Z continued working on the system and knowingly omitted critical limitations during his technical briefing to the defense officials. The situation turned controversial after a demonstration in a controlled military test zone resulted in drones misidentifying civilian figures as hostile entities, drawing public criticism and raising ethical questions.

Analyze the ethical violations committed by Mr. Z in light of the ACM Code of Ethics.

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2. A major urban hospital introduces a deep learning system trained on thousands of radiological images to aid in the detection of early-stage lung cancer. The model, a high-performing convolutional neural network (CNN), was built on years of labeled CT scan data sourced from various institutions. It's reported that diagnostic accuracy rivals that of seasoned radiologists. Encouraged by its performance metrics, the hospital integrates the system into its clinical workflow as a decision support tool. Radiologists are instructed to review the model's risk assessments alongside their evaluations. The hope is that the model will catch subtle anomalies that even experienced eyes might miss.

One day, a 42-year-old patient named Rahim undergoes a routine scan after reporting a persistent cough. The radiologist, Dr. Khan, finds the scan unremarkable—no nodules, no suspicious masses. But the AI model flags it as high-risk, assigning a 91% risk. Rahim's case is discussed in the weekly tumor board meeting. With no clear explanation from the AI system, the oncologist recommends an invasive biopsy, citing the model's high confidence. Dr. Khan protests, arguing against trusting a "black box." The patient's family becomes anxious, torn between medical caution and skepticism of an algorithm with no visible reasoning.

How should the clinical team ethically balance the AI model's high-confidence cancer prediction with the radiologist's contradicting assessment, especially when the model lacks interpretability and the patient's informed consent relies on transparent reasoning?

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3. Nayeem, a final-year student in the Department of Media Studies and Journalism at United International University (UIU), was working on his thesis project titled 'Voices of the Vanishing Land'. The documentary focused on the lives of people affected by riverbank erosion in a village in southern Bangladesh. Over a 2-week field visit, Nayeem conducted interviews, shot visuals of displacement and resilience, and recorded personal stories of families losing their homes to the river. He followed proper ethical procedures by explaining the purpose of the project to the villagers and obtaining informed verbal and written consent and ensured the footage would only be used for academic purposes within a restricted audience of teachers, classmates, and his supervisor.

After sharing a trimmed version of the documentary on his private Facebook profile for classroom peer review, Nayeem discovered that an edited version of his video had been circulated on a popular political Facebook page with over 1.5 million followers. This version included provocative subtitles, dramatic background music, and a misleading caption implying it was part of an anti-government campaign. The video quickly went viral, with local news portals republishing it and social media influencers sharing it widely. Nayeem began receiving threats and hate messages, and the villagers featured in the video grew fearful about being perceived as politically active. Although Nayeem reported the misuse to Facebook and contacted the page, he received no timely response.

In light of this situation, identify two major ethical issues raised by the misuse of Nayeem's video and explain how the principle of informed consent was violated despite his precautions. Additionally, discuss what an appropriate and ethical response from Nayeem should be to protect both himself and the villagers involved.

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4. A mobile fitness app offers personalized workout routines by tracking users' activity levels and locations throughout the day. The app collects GPS data, heart rate, and time spent at specific locations (e.g., gyms, parks). This data is also shared with third-party advertisers to suggest fitness products based on the user's movement and shopping patterns. The data collection and sharing terms are buried deep in the app's terms and conditions, which users often accept without reading.

Is it ethical for the app to collect and share user data with third parties based on vague consent buried in terms and conditions? What practices should the app follow to ensure ethical data usage and protect user privacy? Prepare a sample consent letter for this purpose.

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5. In the heart of Hell's Kitchen, Wilson Fisk re-emerges under the banner of environmental reform. Through his company, he launched the 'Green Hell's Kitchen Initiative', promising to transform the rundown neighborhood into New York's first "zero-emissions zone." The plan includes AI-controlled waste disposal, high-efficiency housing, and underground power plants running on experimental geothermal tech.

On the surface, the project appears to be a green revolution. Waste levels drop. Air quality improves. Fisk receives praise from city officials. But investigative reports by Karen Page reveal deeper problems: the geothermal units are leaking toxins into the bedrock, harming the underground water table. The construction displaced hundreds of low-income families, and the initiative is importing cheap "eco-materials" from overseas factories with severe pollution records. Matt Murdock investigates further and uncovers that the project is a front to launder Fisk's operations while gaining public approval. He's torn; the project is making the city cleaner, but at what cost?

As an ethically minded technologist, how would you respond to this situation? Is it acceptable to pursue environmental goals through methods that harm ecosystems elsewhere or displace communities? Give support in favour of your answer.

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